Quant Club, BITS Pilani

Research Case Study | Portfolio Construction

Multi-Strategy Portfolio Allocation

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Dynamic risk budgeting across MTUM / USMV / VIXY | Sample: 2019-01-03 to 2026-08-05 (1907 trading days)

1. Executive Summary

This case study evaluates how a multi-strategy allocator should distribute capital across three liquid strategy sleeves - momentum (MTUM), defensive min-vol (USMV), and crisis/vol (VIXY) - using a 60-day rolling risk engine. Net of 10bps transaction costs, inverse-volatility allocation delivered 7.8% annualized return and Sharpe 0.30, versus equal-weight at -4.6% and Sharpe -0.50. A static 60/40 (MTUM/USMV) earned the highest return (15.3%) by avoiding VIXY's long-run roll decay. Recommendation: use inverse-vol when the mandate is risk-adjusted capital preservation with a crisis sleeve; use 60/40 when the mandate is return-seeking and hedges are optional.

2. Investment Thesis

Multi-strategy hedge funds allocate capital across strategy return streams whose risk contributions change with market regimes. The core problem is dynamic risk budgeting: when one sleeve's volatility spikes, capital should shrink there and rotate toward more stable sleeves - without abandoning diversification. This study treats publicly traded ETFs as transparent proxies for momentum, defensive, and crisis strategies, then compares naive inverse-vol risk parity against covariance-aware minimum variance and max diversification, all versus static benchmarks.

3. Strategy Sleeves

Momentum (MTUM): equity factor tilt that participates in trending markets; higher beta and drawdown risk in abrupt reversals. Defensive / Min-Vol (USMV): low-beta sleeve oriented to capital preservation in risk-off environments; steadier path, lower upside in strong bulls. Crisis / Vol (VIXY): short-term VIX futures ETF that tends to rise when equity vol spikes; provides convexity in crashes but suffers structural roll/contango decay in calm markets. That decay is the central economic trade-off in this portfolio.

4. Methodology

Risk window: trailing 60 trading days for annualized volatility and the covariance matrix. No lookahead - weights for day t use information through t-1 only. Inverse-vol: weight_i proportional to 1/sigma_i, renormalized to 100% invested (long-only, no leverage). Minimum variance: minimize portfolio variance w'Sigma w with sleeve bounds 5%-60%. Max diversification: maximize (w'sigma) / sqrt(w'Sigma w) under the same bounds. Friction: 10bps per unit of one-way turnover on dynamic methods. Risk-free rate for Sharpe/Sortino: 4.5%.

5. Results (Net of Costs)

Method	Ann Ret	Sharpe	Max DD	VaR 95%
InvVol	7.8%	0.30	-22.8%	-0.9%
MinVar	5.0%	0.05	-28.0%	-0.8%
MaxDiv	3.8%	-0.07	-36.6%	-0.9%
EqualWeight	-4.6%	-0.50	-57.0%	-1.5%
60/40	15.3%	0.56	-33.6%	-1.7%

InvVol cut max drawdown to -22.8% vs -57.0% for equal-weight, and lifted Sharpe from -0.50 to 0.30. MinVar was more conservative (vol 9.8%) but lagged InvVol on risk-adjusted return after costs. 60/40 won on absolute return because it holds zero VIXY. InvVol 1-day 95% historical VaR: -0.9%.

6. Regime Analysis

COVID crash (Feb-Mar 2020): equal-weight's VIXY sleeve produced a large positive period return (39.1%), illustrating crisis convexity; 60/40 fell -21.8% without a vol hedge. InvVol (-5.9%) sat between pure risk-on and equal-weight's heavy VIXY bet. 2022 rate shock: all sleeves struggled; InvVol returned -13.8% vs -14.5% for

60/40. In a grinding rates-driven bear, the crisis sleeve helped less than in March 2020 - an important mandate implication.

7. Risks and Limitations

VIXY is a futures-based product with roll/contango costs; it is a poor permanent hedge and a better episodic crisis sleeve. Equal-weight's long-run underperformance is largely this drag. ETF proxies are not live hedge-fund P&L; they approximate strategy style exposures only. Daily rebalancing inflates turnover versus institutional monthly/quarterly practice (InvVol ann. turnover ~1.8x). Long-only, fully invested, no leverage, no shorting, no cash sleeve. Past regime performance is not a forecast; 60-day windows can lag abrupt regime shifts.

8. Conclusion

For a risk-conscious allocator who wants a dedicated crisis sleeve, inverse-volatility risk parity improves Sharpe and drawdowns versus naive equal-weight after realistic friction. For a return-seeking mandate that can live without permanent vol exposure, static 60/40 (momentum + min-vol) dominates on CAGR because it avoids VIXY decay. The professional takeaway is not that one method always wins - it is that the optimal allocator depends on whether the investment committee values crisis convexity enough to pay its carry cost.

Not financial advice. Research / educational case study only. See STRATEGY_RATIONALE.md, EXHIBIT_CAPTIONS.md, and INTERVIEW_PREP.md.